

Problem 1: A key changes each letter while keeping its place in a message. Use key P below to encode all three messages. Every output letter appears once in the key, so you can also follow it backward.

Input	A	B	C	D
P output	B	C	A	D

Message	Encoded message
ABBA	
ADAD	
ABCD	

Problem 2: Decode CAAC and DBDB. Check each result by encoding it again.

Problem 3: Invent a key using A,B,C,D once each as outputs. Encode ABBA. Compare it with P’s output. Mark the positions that match each other in both outputs.

Input	A	B	C	D
My output				

Problem 4: Start with ABCD. Change every letter at once using P. Apply P to the new message, and keep going. Record each row. Circle the first turn after 0 when the entire row returns to ABCD. Do the same with Q.

Input	A	B	C	D
P	B	C	A	D
Q	B	A	D	C

Turn	P message	Q message
0	ABCD	ABCD
1		
2		
3		
4		

Problem 5: In each key, follow A until it comes back. Draw arrows between the letters it visits, closing the loop. Choose a letter not yet drawn and repeat. A letter that stays put gets an arrow back to itself. Draw all the loops for P and Q.

Problem 6: Make a key that takes four turns to return the whole message ABCD. Build it with cards, draw its arrows, then test all four turns.

Problem 7: Use four letter cards to build each arrangement below. An arrow loop must return to its own start. For each arrangement, test the first turn when all four cards are home. Write your key or draw the loops.

Loop sizes	My key or loops	First return
3 + 1		
2 + 2		
2 + 1 + 1		
1 + 1 + 1 + 1		

Problem 8: Compare these four arrangements with your four-turn key. Try to build a six-turn key with four letters. Then add E and try a loop of three letters and a loop of two. Record what happens.